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INSTITUTE OF ENGINEERING

SAGARMATHA ENGINEERING COLLEGE

A

MINOR PROJECT REPORT

ON

STAY

BY

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A PROJECT REPORT SUBMITTED TO THE DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR IN COMPUTER ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

SANEPA, LALITPUR, NEPAL

OCT 2020

DECLARATION

We at this moment declare that the report of the project work titled as "STAY" which is being submitted to the Sagarmatha Engineering College, IOE, Tribhuvan University, in the partial fulfillment of the requirements for the Degree of Bachelor of Engineering in Computer Engineering, Which has been carried out by us. The material contained in this report has not been submitted to any University or Institution for the award of any degree.

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ABSTRACT

We propose to rented room system with web application for users.A majority of unpleasant issues faced by landlords and tenants seem to grow due to lack of communication. Late rent payments turns into an eviction. A non-functional door or a broken window becomes a reason to break a lease. Fortunately, building a positive relationship doesn't take much effort and it all starts with one basic idea: communication. Stay web app provides the solution by offering the ability to share issues between tenants and landlords which leads to rapid issue resolution and fewer misunderstanding. The goal of the web app is to create a better relationship between tenants and landlord. The stay web app is designed to support both tenants and landlords by enabling them to document and communicate . It helps the tenant with making Payments and it also helps the Landlord to keep track of issues posted by tenants. Effectively resolving the apartment issues is important to the tenant's longterm future and Stay web app will be an important tool for creating rental housing stability by helping tenants speak with greater credibility through initiating and documenting communications and building productive relationships with landlords.

Keywords: Room Renting Nepal,Open Street Map, DBMS(Database Management System), Django backend

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LIST OF ABBREVIATIONS

AI	: Artificial Intelligence
DFD	: Data Flow Diagram
DBMS	: Database Management System

CHAPTER 1

INTRODUCTION

Rental room management has become important factor in modern society hence the need to have a rental house management system. This chapter will provide a brief understanding about background of study, definition of the project problem statement, its objectives, scopes, project justification, risks, project deliverables . Housing has a central importance to quality of life with considerable economic, social, cultural and personal significance.

1.1 BACKGROUND INTRODUCTION

This project "STAY" is progressive web application designed to address the problem of room finding for students and families who have come with bigger dreams and for other different kinds of opportunities to make their life better.This web application is designed to connect tenant directly to houseowner. It helps tenants to save their time instead of roaming around for room hunting which may take days to get the desired room.

1.2 MOTIVATION

Project Motive

Rental room management system has become important factor in modern society hence the need to have a rental house management web application.

It is difficult to find house for the renter in a specific area.

It is also difficult to find the renter on time, for the landlord and property managers.

Resently This Kinds of system working in Kathmandu Valley.

1.3 PROBLEM STATEMENT

In our day-to-day life, buyer's face a lot of issues related to household such as reporting an issue, tracking the location and receiving necessary information instantly etc.., and similarly seller's also deal with issues such as keeping track of issues in apartments, notifying that a package has arrived, sending emergency information quickly to the buyers etc. Currently there is no app which focuses on to resolve this particular issue. The existing app help to solve the issue but not in an effective manner.

1.4 PROBLEM SOLUTION

To help overcome the problems faced by buyers and sellers, a new app is developed. This app focuses on to build a better relationship between buyers and sellers by simplifying many tasks such as sending automatic rent reminders, package notifications, utilities bill, emergency info, location information, documents and sharing issues. Explain how your project will solve existing problems

1.5 OBJECTIVE

This research work is conducted in order to:

- To develop a rental house management system that allows the user to view customers data as well as houses record
- To study and analyze the requirement specifications of the rental house management system

1.6 SCOPE AND APPLICATION

Scope of 'STAY' is to provide user friendly,mobile friendly,simple,interactive and responsive design as the world gets more digital. This project also has feature of map pinned location where you can also know the house where it is located.Eiditinr is done by admin for security purpose.

CHAPTER 2

LITERATURE REVIEW

Literature Review

In this world, where everyone works towards betterment of their life; here, in Nepal too, people from all over the country migrate to urban areas for various reasons for better education, for better job opportunities, for business purposes, etc. Nowadays where everything is being digitized around the globe, we are also trying to digitize a minor aspect of many people's life i.e. room finding. We are creating 'STAY' to help those people to find a place to stay.As we were roaming through the internet to look for any such projects that have been made, we found out 'roomgo'.

2.1 ROOMGO

roomgo[1] is a company which connects people seeking shared living spaces and provides unique accommodation

2.2 GHARBETI.COM

'gharbeti.com[2]' has the most simplest way of putting advertisement without the need of a broker. It also shows all the details of the room or property and has a option of applying and the owner contacting the possible renters as per their liking.

2.3 HAMROBAZAR.COM

'hamrobazar.com[3]' rather than only being land property, renting application is a fully fledged application in which people can keep advertisement and give their contact information and communicate on chat box in public for enquiry or through the given phone number of the owner.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

METHODOLOGY

Working Principle

STAY is web application in which two different forms of user i.e. house owner and tenant interacts to the system in order to meet their own requirements.To develop the system,we used different techniques which make thw user interaction in the system with greater flexibility and influence. Authentication is the main part of the system which doesn't give access to unauthorized users. This project used postgre sql i.e. database to store data.This project used open street map i.e. open source and individual buildings are mapped which helps us to pin location of the house.Here house owner gives the location of house,upload the pictures of house and monthly rent of the house inorder to do so one must sign up.After all information is uploaded which are necessary for tenant,all information is stored in database.After successful login in the system by tenant,homepage consist of two option choose partner and search room will be appear in which they can select any option then tenant can access the information uploaded by the house woner and book any house that is suitable for him.Tenant can search to any specific area which is favourable for them and their day to day work for location tracking we use open street map.

3.1 PROJECT DEVELOPMENT LIFE-CYCLE MODEL

A process model for software engineering is chosen based on the nature of the project and application, the methods and tools to be used, and the controls and deliverance's that are required. According to the nature of this project we have used Agile model[1] for development. Agile Model[1] is a method of software development where the product is designed, implemented and tested in stages thus making adding new features easier when ever new feature is needed to be added.

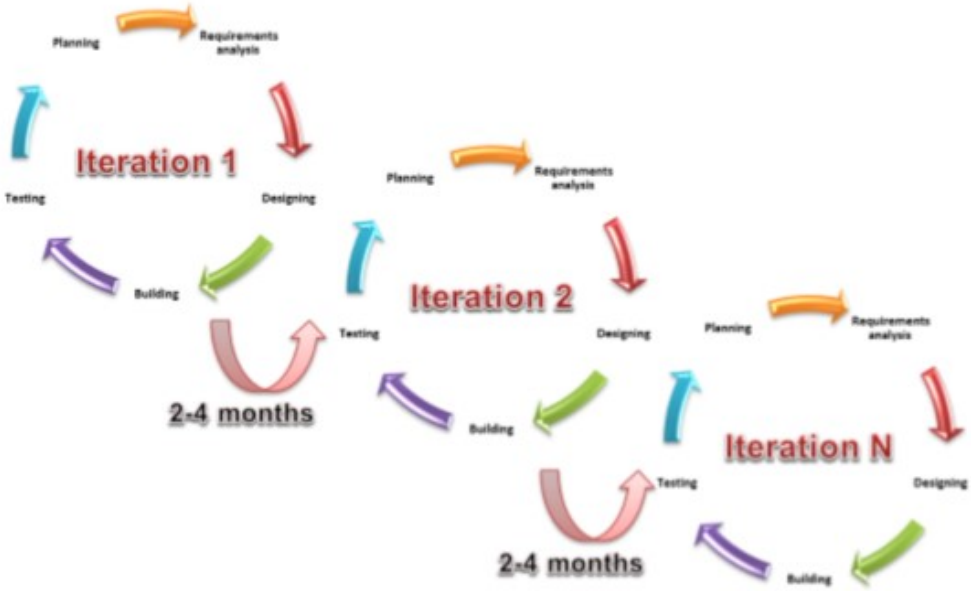


Figure 3.1: AGILE MODEL

3.2 REQUIREMENT ANALYSIS

3.2.1 FUNCTIONAL REQUIREMENT

3.2.2 NON FUNCTIONAL REQUIREMENT

3.3 SYSTEM DESIGN

3.3.1 UML DESIGN

UML design is the shortest form of "Unified Modeling Language". The purpose of this modelling language is to visualize the design of the system. There are several kinds of UML diagrams but here we use only Class diagram:

Class Diagram: This is the most used UML diagram in the field of software engineering design. It is called as a main building block of any object-oriented solution. Usually it illustrates the classes in a system, attributes and operations of each class and also the relationship between each class. Below is the "CLASS DIAGRAM" of online room renting system "STAY"

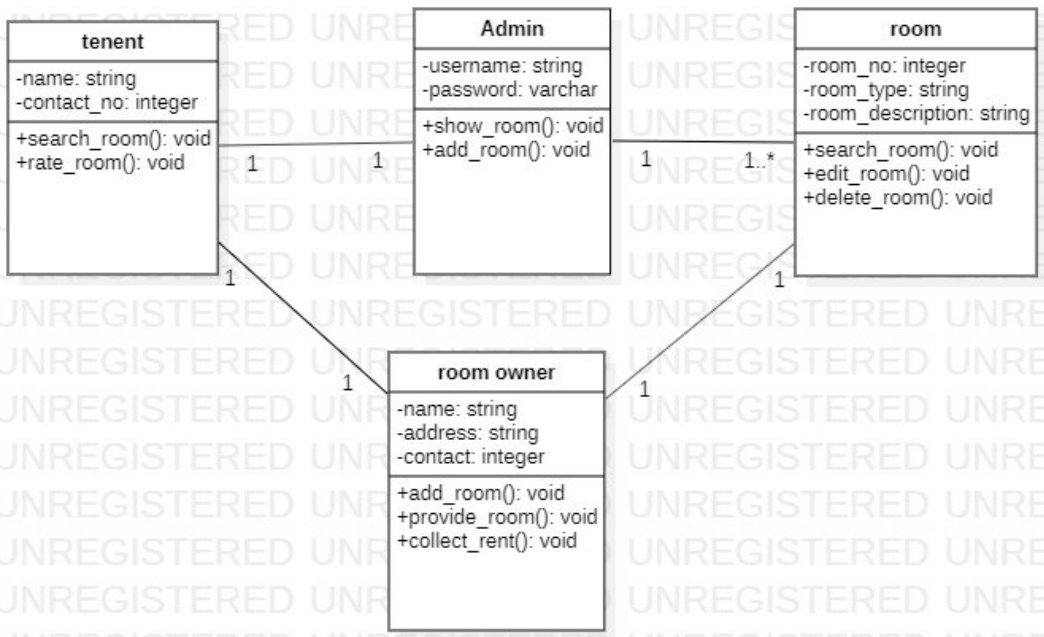


Figure 3.2: CLASS DIAGRAM

3.3.2 DATA FLOW DIAGRAM

A data flow diagram is a graphical representation that depicts information flow and the transforms that are applied as data move from input to output. The basic form of a data flow diagram, also known as a data flow graph or a bubble chart. DFD is an abstract description of the system. The data flow diagram may be used to represent a system or software at any level of abstraction. DFDs may be partitioned into levels that represent increasing information flow and functional detail. Therefore, the DFD provides a mechanism for functional modeling 9 as well as information flow modeling. DFDs are very useful in understanding a system and can be effectively used during analysis. DFDs can be hierarchically organized, which helps in progressively partitioning and analyzing large systems. Such DFDs are called leveled DFDs. Context diagram is a diagram in which the entire system is treated as a single process and all its inputs, outputs, sinks, and sources are identified and shown.

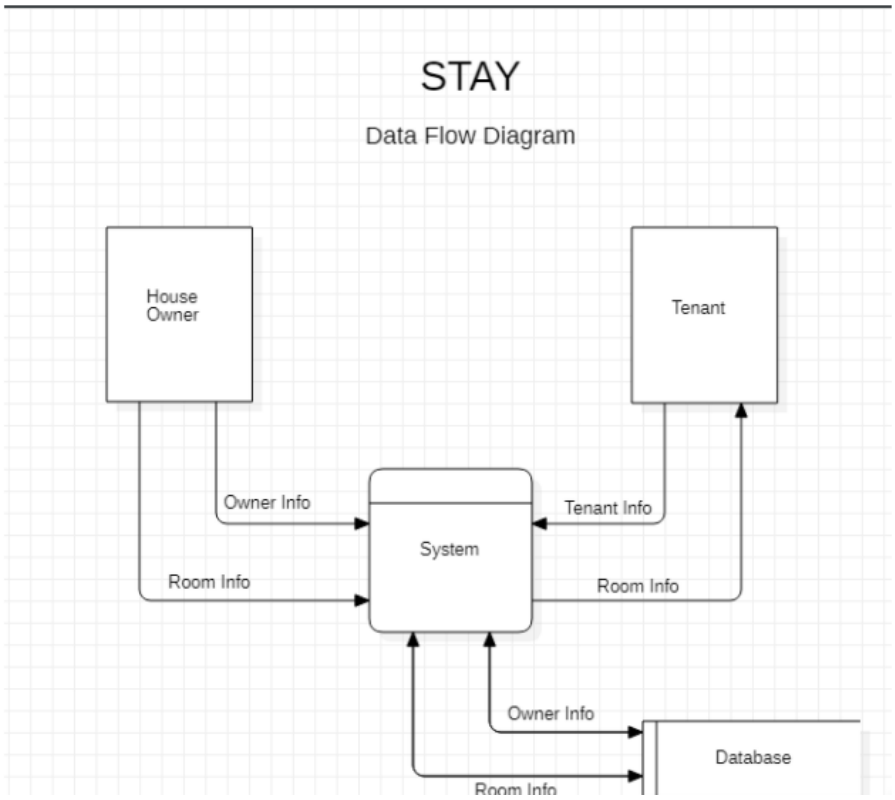


Figure 3.3: DFD LEVEL 0

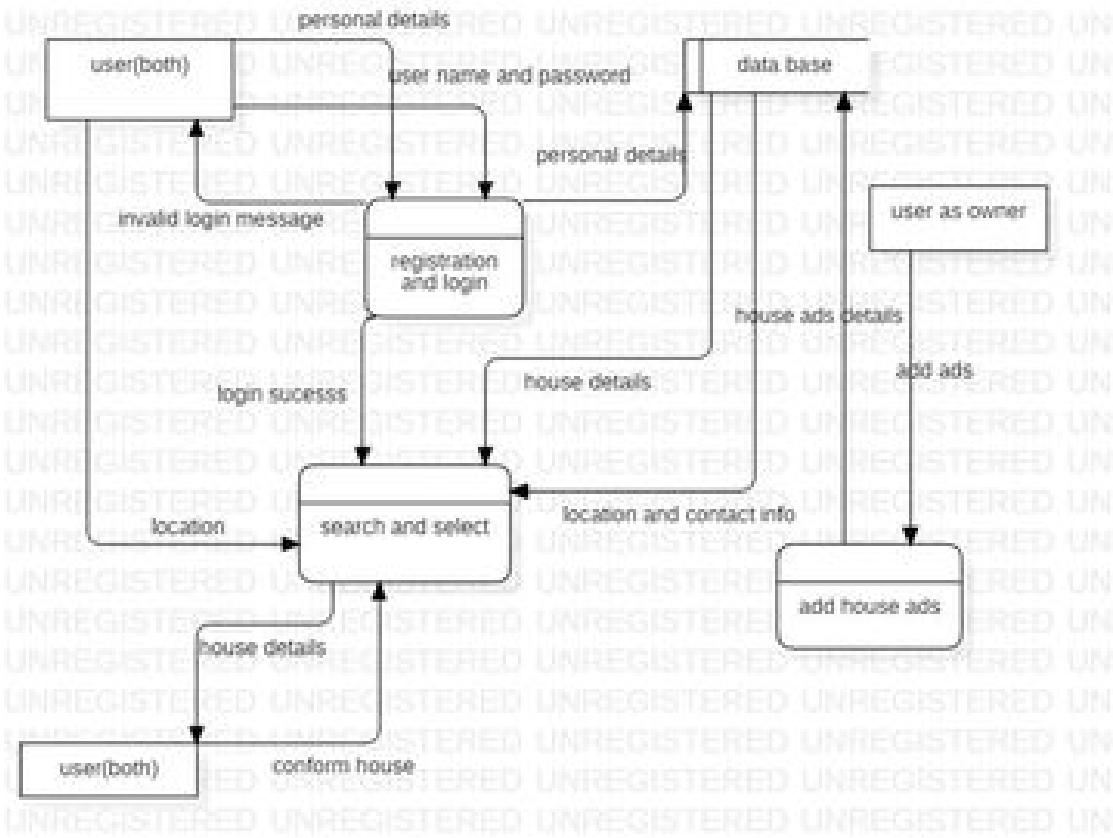


Figure 3.4: DFD LEVEL 1

CHAPTER 4

IMPLEMENTATION DETAILS

Implementation of tools: This project is implemented on android using html, css , bootstrap , js for frontend and anaconda virtual environment ,open street map i.e. open source and individual buildings are mapped which helps us to pin location,django is used in backend.postgre sql is used for data base inorder to store the data of user whose detail description is given below:-

4.1 HTML

Hypertext Markup Language (HTML) is the standard markup language for documents designed to be displayed in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript.

4.2 CSS

CSS is the language for describing the presentation of Web pages, including colors, layout, and fonts. It allows one to adapt the presentation to different types of devices, such as large screens, small screens, or printers. CSS is independent of HTML and can be used with any XML-based markup language.

4.3 BOOTSTRAP

Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first front-end web development. It contains CSS- and (optionally) JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components.

4.4 JAVA SCRIPT

JavaScript is a text-based programming language used both on the client-side and server-side that allows you to make web pages interactive. Where HTML and CSS are languages that give structure and style to web pages, JavaScript gives web pages interactive elements that engage a user.

4.5 DJANGO

Django is a high-level Python web framework that enables rapid development of secure and maintainable websites. Built by experienced developers, Django takes care of much of the hassle of web development, so you can focus on writing your app without needing to reinvent the wheel

4.6 ANACONDA

Anaconda is a conditional free and open-source distribution of the Python and R programming languages for scientific computing (data science, machine learning applications, large-scale data processing, predictive analytics, etc.), that aims to simplify package management and deployment.

4.7 OPENSTREET MAP

OpenStreetMap is a collaborative project to create a free editable map of the world. The geodata underlying the map is considered the primary output of the project.

4.8 POSTGRE SQL

PostgreSQL is a powerful, open source object-relational database system that uses and extends the SQL language combined with many features that safely store and scale the most complicated data workloads.

STAY

Help Sign up Login List your Room

Get Rooms and Room-mates

Easily . Safely . Securely

Get a roommate

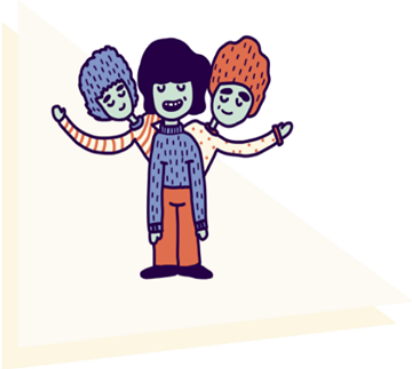
Find roommates you want to live with.

List your room

Get a room

Find rooms for rent in 3 steps.
Search, Chat, Move

Find your room



Rent on your terms

It's your place. Rent it how you want.

DISCOVER ROOMMATE

Use STAY if you're looking for a new friend to move in & split the rent.

TERMS AND CONDITIONS

Customize renting terms on who your tenant/roommate should be like.

Setup your house rules and list your room for rent.

Rent Easier Live Secure

STAY makes the entire roommate finding experience simple. Everything is secured and documented.

ONLINE BOOKING

Easily review booking requests from interested renters. Once you approve their request to book your room, we'll take your listing off the market and set up secure payment.

SECURED CONTACTS

When potential tenants are interested in your post. You can review their profile and only after your interest contacts are visible to both parties.

Testimonials

Overall it was a successful hunt. Quick and easy apartment hunt. Lots of tools to help you get it done as efficiently as possible. Overall it was a successful hunt.

Bibek Thakuri

Overall it was a successful hunt. Quick and easy apartment hunt. Lots of tools to help you get it done as efficiently as possible. Overall it was a successful hunt.

Kartabya Aryal

Overall it was a successful hunt. Quick and easy apartment hunt. Lots of tools to help you get it done as efficiently as possible. Overall it was a successful hunt.

Impress Bashyal

Why STAY works?

At STAY, our mission is to help you rent a room easily, safely & securely. There are many options to find rooms and flats for rent but when it comes to renting a room or finding roommates, STAY is the best roommate finder in Nepal.

Whether you are looking for a room for rent near you or a flat for rent, your personalized search results will always show you the best rooms for rent in houses based on your lifestyle preferences, lease terms, pricing and much more.

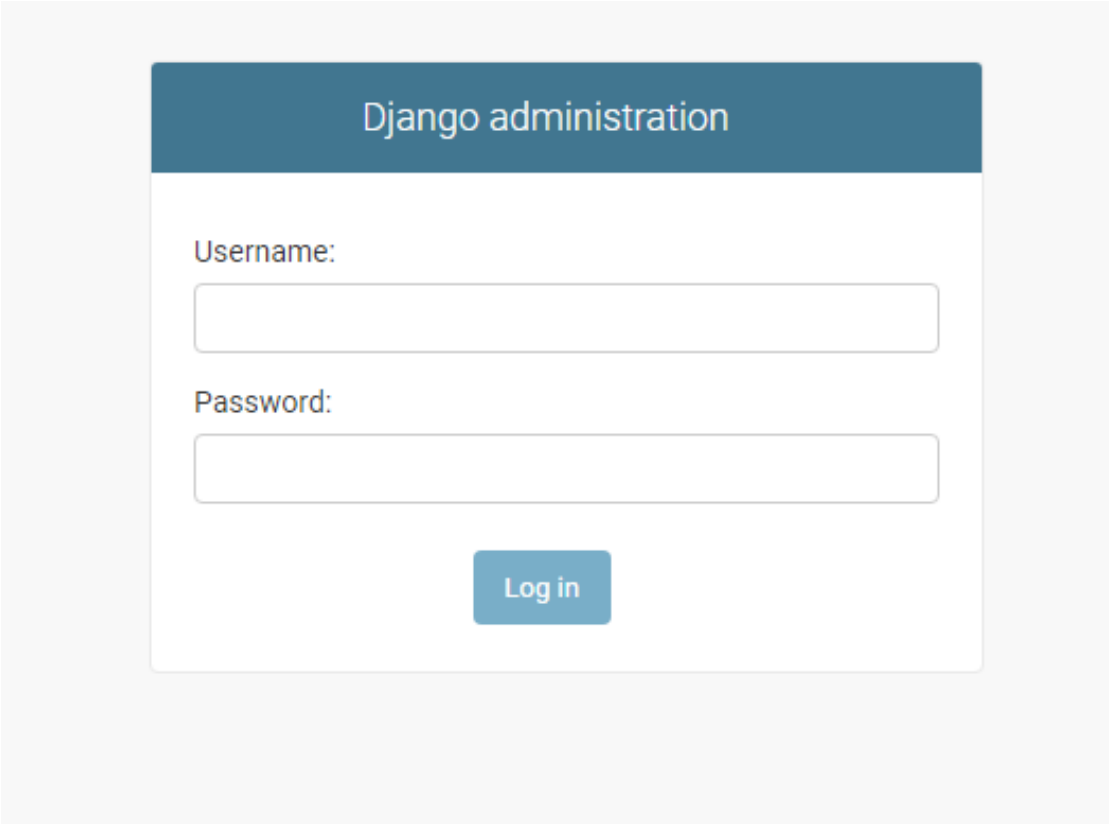


Figure 4.2: ADMIN LOGIN PAGE

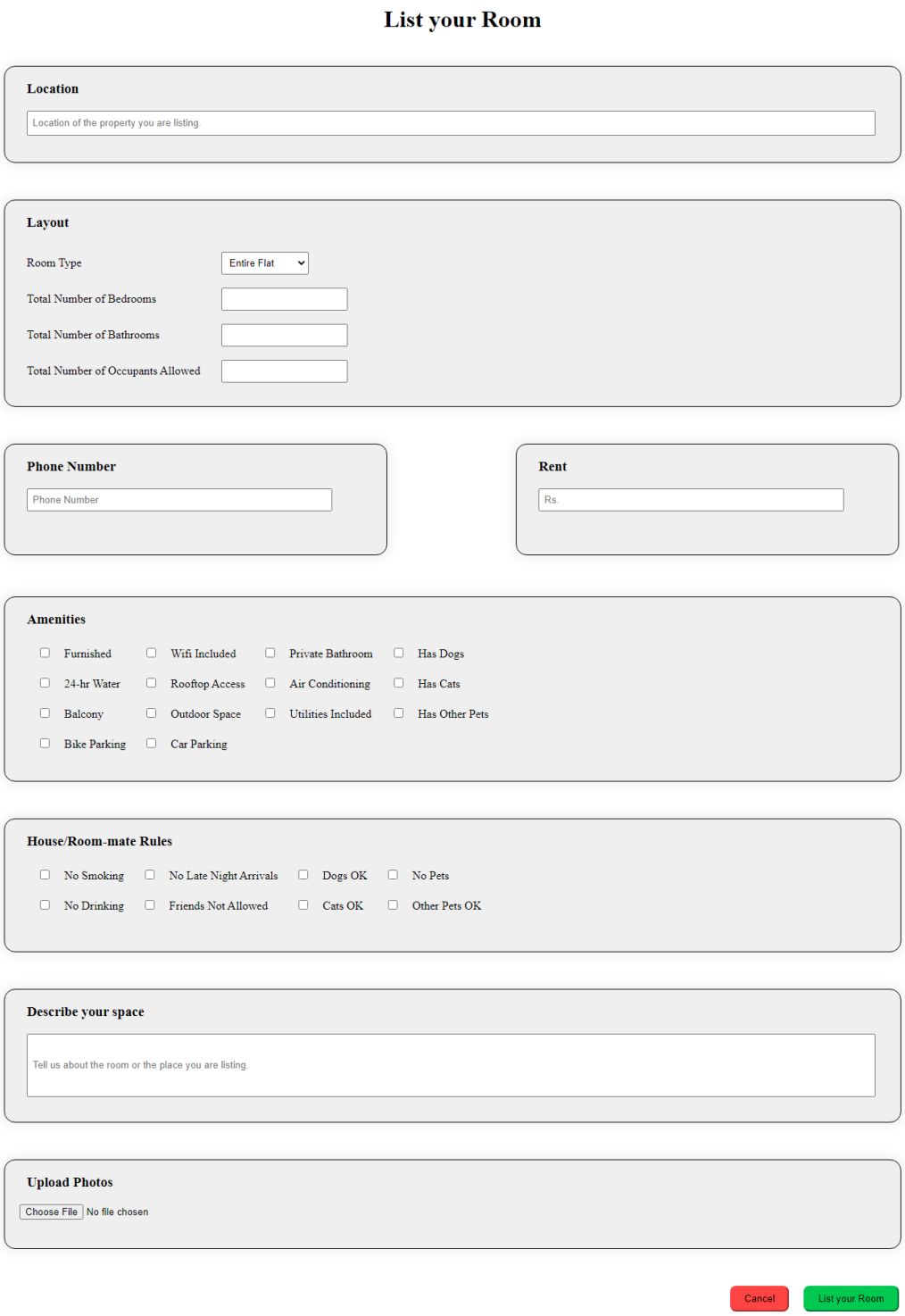


Figure 4.3: ROOM LISTING PAGE

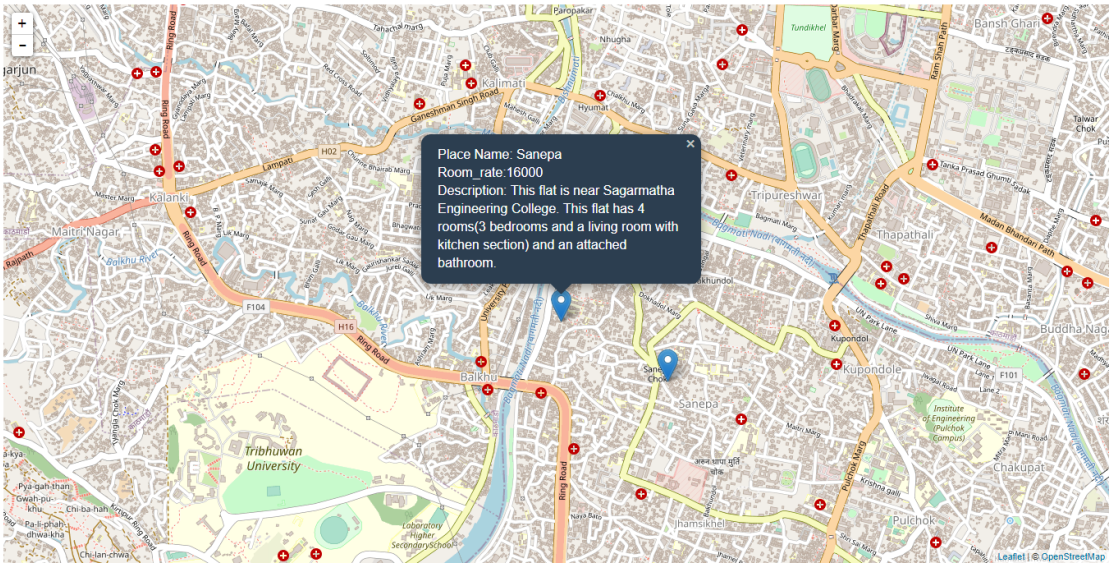


Figure 4.4: PINNED LOCATION IN MAP

CHAPTER 5

EVALUATION AND RESULT

The overall system was tested for any faults and errors and the final result or output was analyzed. Firstly, the UI of the system was interactive for better user experience. Another Part is registration of user, there are certain compulsory field in registration form. If all the compulsory field are entered, then the successful registration is done else the registration is invalid or not completed. Moreover, each field consists of varied form of attributes, all the field should be filled as per the desired attributes for successful registration One of the most important part of the project is use of open street map. The information of users is registered or stored in firebase database.If a user want a room then ,tenant must register in the STAY then search the desired room.If the room is available then tenant should must get discussed with room owner.if user met the room owner criteria then tenant may shift in the room.

CHAPTER 6

EPILOUGE

6.1 CONCLUSION

With the completion of this project, the main aim of the project is achieved, that is easy and efficient way of finding the rooms. This application integrates various functionalities to provide the best rooms with good room owner making their work an efficient, time-saving and less tedious. With the end of the project, the members involved in project gained a lot of experience on team work and they discovered various predicted and unpredicted problems and also implemented various idea to solve them. Various resources like video tutorials, text tutorials, internet and learning materials were used to make project complete.

6.2 LIMITATION

The project assists well to record the problem related to transfusion activity. However, this project has some limitations. Such as notification of room available is sent to the registered user and also the app consists of many vacant room available.

6.3 FUTURE ENHANCEMENT

To further enhance the capability of this application, the following recommended features is to be incorporated into the system:
Notification of room available in desired location with well facilities.
List of many rooms all over the nepal.

REFERENCES

ROOMGO[1] https://www.roomgo.net/

GHARBETI[2] http://gharbeti.com/

HAMROBAZAR[3] https://hamrobazaar.com/